A Multi-Hazard Assessment of 2,696 US Data Centers

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Aspiring Scientists Summer Internship Program 2026

Background

- AI and cloud demand has concentrated US computing into a few dozen metro clusters that face earthquakes, wildfire and flooding.
- Existing sources are proprietary, or give city-level coordinates never checked against a building.
- We built a free, public, building-level inventory of the contiguous US, and tested how much the answer moves when a coordinate is wrong.
- We measured exposure, meaning whether a site lies in a hazard zone. We did not model damage, so nothing here is a loss estimate.

Limitations

- Coverage is uneven. 83% of our Virginia sites come from OpenStreetMap against 48% of our California ones, so part of the gap between states reflects who did the mapping.
- Hail and wind come from eyewitness reports, so within a state their rates track the number of other sites within 40 km (Spearman 0.39 and 0.34). We left both out, along with lightning, tornado and hurricane.
- No public source lists power capacity, so a server room counts the same as a 100 MW campus.
- 40 sites shown as clear sit outside FEMA's mapped extent and 8 outside the wildfire raster, so 35% is a floor.

Materials and Methods

We merged OpenStreetMap, PeeringDB and Wikidata into one record per site across the contiguous US, matching on name and distance, with written evidence kept for all 194 merges.

Against FEMA USA Structures footprints, 1,681 coordinates fall inside a building, 993 match the nearest and 22 have no match. A second non-OSM source confirmed 1,546, and we re-checked the 1,150 least certain with an AI-assisted pipeline we then reviewed.

Our first seismic layer failed validation against USGS reference values (22 of 150 sites within 10%), so we replaced it:

Earthquake	USGS ASCE 7-22 shaking, 2% chance in 50 yr
Wildfire	USFS Hazard Potential 270 m, max in 2.4 km
Flood	FEMA flood layer, 1-in-100-year floodplain
Water stress	WRI Aqueduct 4.0, river-basin scale
Uncertainty	500 relocations per site, 10-500 m by precision

A site counts as exposed if it lies within 2.4 km of land rated High or Very High for wildfire, inside a FEMA Special Flood Hazard Area, or at peak ground acceleration of 0.30 g or above. Wildfire is buffered because 2,284 sites sit on non-burnable land, where the pixel under the building says nothing.

Results

2,696	954	952	563
facilities located	face a mapped hazard	in a high water-stress basin	water-stressed only

Hazard and water stress pick out largely different buildings. Counting both raises the share under a physical constraint from 35% to 56%.

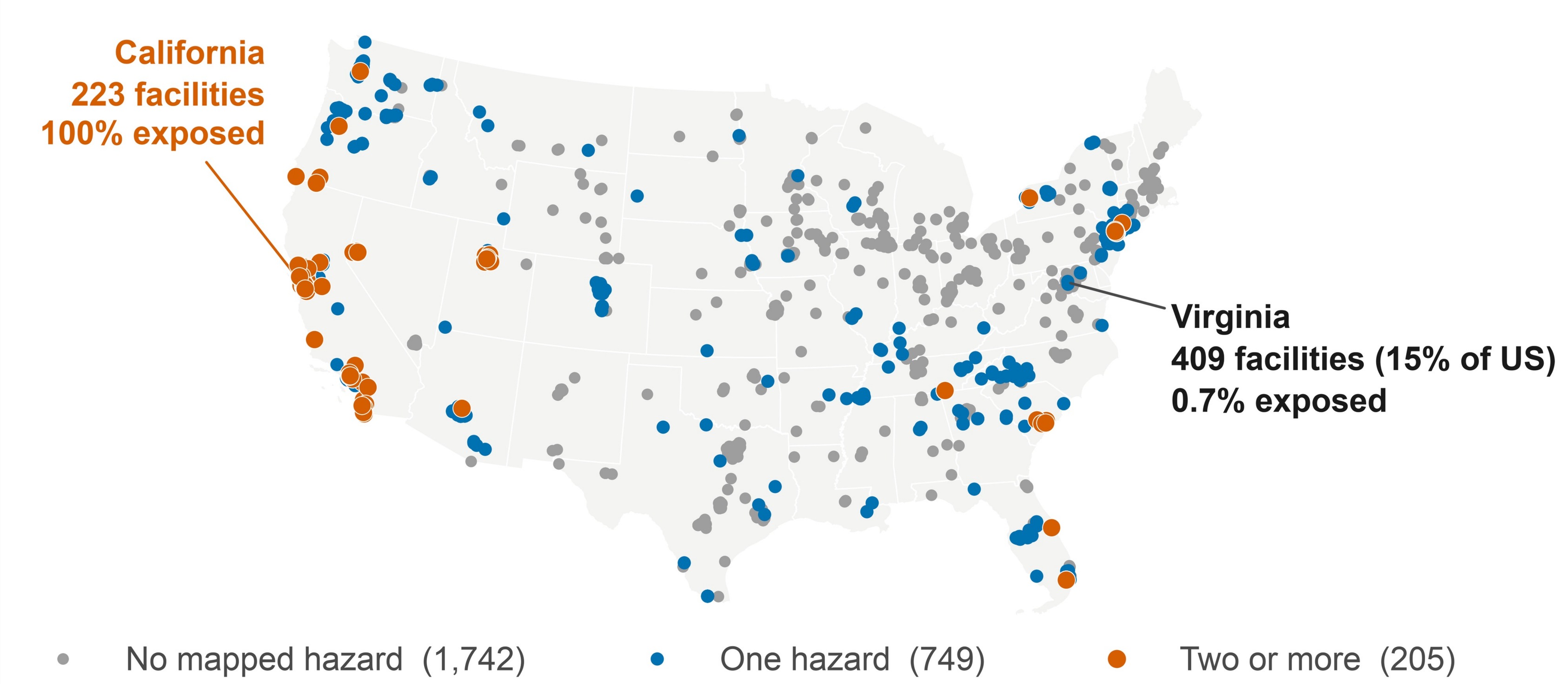

Fig 1. One third of US data centers face a mapped hazard. Of the 205 facing two or more (orange), California holds 119, with smaller clusters in Utah, Nevada and New Jersey.

952 sites lie in a high or extremely-high water-stress basin, almost exactly the 954 facing a mapped hazard, but only 389 are the same sites. Some of the largest clusters are stressed while their hazard maps read clear: Virginia is 0.7% hazard-exposed and 31% water-stressed, Illinois 2.7% and 95%.

Which hazard dominates depends on the region. New Jersey reaches 100% through wildfire alone and 0% through earthquake, California 96% earthquake. Nationally 646 sites are exposed to wildfire, 448 to earthquake and 71 to flood.

Widening the wildfire buffer from 2.4 km to 5 km takes the wildfire count from 646 to 953, so that result is a statement about the buffer as much as about the site.

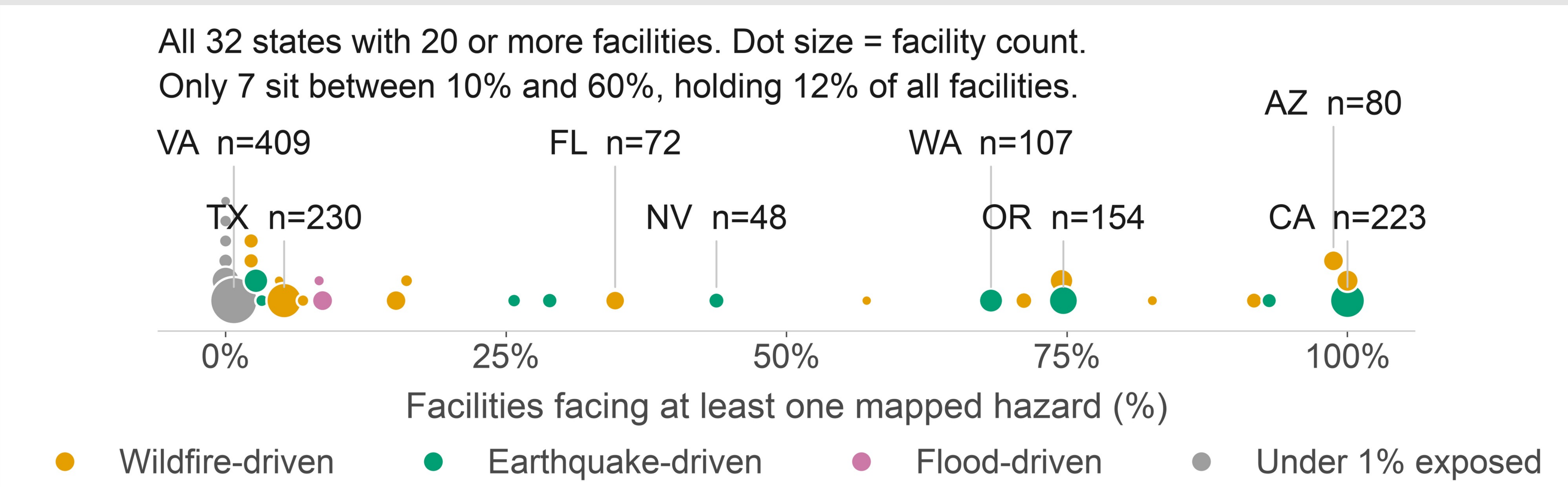

Fig 2. All 32 states with 20 or more sites, coloured by dominant hazard.

Conclusions

- A single national rate is close to meaningless. Only 7 of 32 states with 20 or more sites fall between 10% and 60% exposed.
- We did not measure any site's water use, but the basins under the largest clusters are already over-drawn, and a hazard screen misses them.
- New Jersey and California are both 100% exposed for opposite reasons, so one score hides what a site faces.
- Next: fragility curves, to turn exposure into loss.

Two robustness checks

Sampling across a whole footprint instead of its centre changed the answer for 29 of 326 comparable sites. Taller buildings looked more exposed nationally, but that vanished within states.

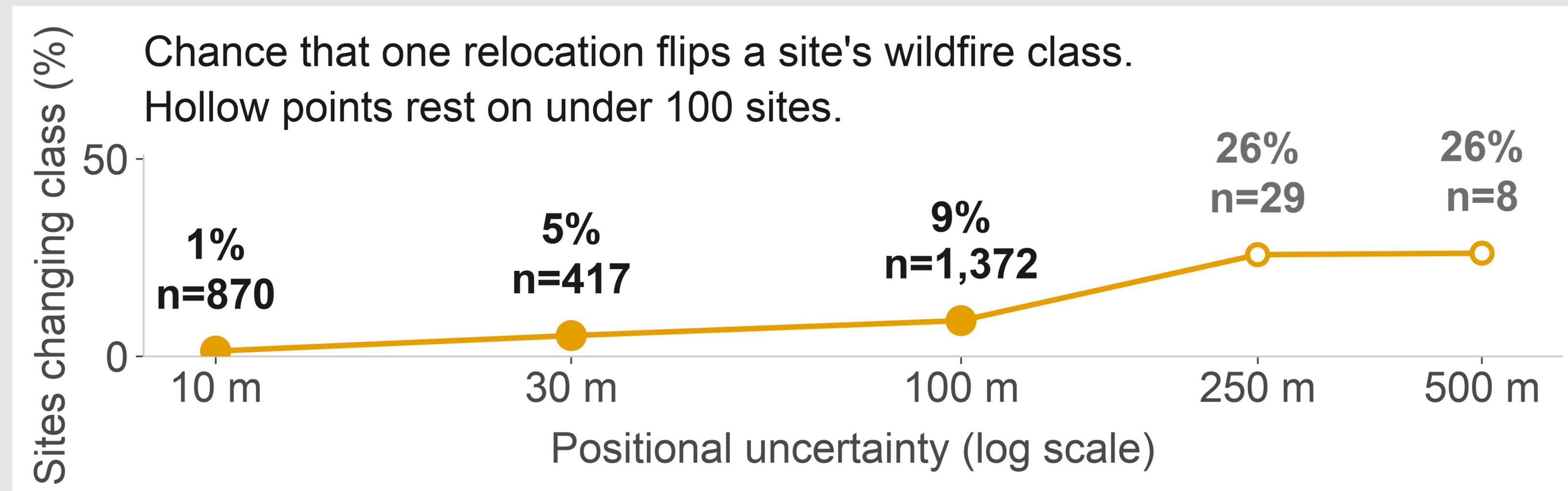

Fig 3. Wildfire class under coordinate error.

Major Citations

Dillon, G.K. (2023) Wildfire Hazard Potential for the United States, 270-m, v2023. USDA Forest Service. doi:10.2737/RDS-2015-0047-4

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FEMA and ORNL (2024) USA Structures. gis.fema.gov

WRI (2023) Aqueduct 4.0 Water Risk Atlas.

Acknowledgements

This research was made possible through the support of George Mason University's College of Science, which supports the ASSIP Program, and by the NCAR multi-hazards project.